SYNTHETIC DATA — 교육용 가상 데이터이며 실제 제품 사양·운영 기준과 무관합니다.

Cast Resin Dry-Type TRF 630kVA (TRF-DRY-630) — Datasheet

1. Overview

Cast resin dry-type transformer 630kVA, 11/0.415kV, Dyn11, AN/AF. Product family: Power transformers. Lifecycle status: **ACTIVE**.

2. Technical specifications

2.1. Ratings

Item	Value
Rated power	630 kVA (AN) / 880 kVA (AF, with fans)
Voltage ratio	11 kV / 0.415 kV
Vector group	Dyn11

2.2. Performance and characteristics

Item	Value
Impedance voltage	6% at 120 °C
Off-circuit tap range	±2 x 2.5%
Insulation class / temperature rise	Class F / 100 K

2.3. Endurance and communication

Item	Value
No-load loss	1,300 W
Load loss at 120 °C	6,500 W

2.4. Protection and options

Item	Value
Environmental / climatic / fire class	E2 / C2 / F1

2.5. Monitoring and auxiliaries

Item	Value
Winding temperature alarm / trip	140 °C / 155 °C (PT100)

3. Installation data

3.1. Terminals

- Dimensions (W x H x D): 1500 x 900 x 1650 mm
- Weight: 1850 kg
- Degree of protection: IP00 (IP21/IP23 enclosure option)
- Operating temperature: -25 to +40 °C

4. Standards and certifications

Applicable standards:

- IEC 60076-11

Certifications held:

- Type test report (IEC 60076-11)
- KC

Certificates and declarations of conformity are available from the Certification & Compliance team on request. Certifications not listed above are not held for this product; project-specific evaluation can be requested.

5. Ordering information

Item	Value
Ordering code	TRF-DRY-630
Standard lead time	10 weeks ex-works
Minimum order quantity	1
Warranty	24 months (see POL-WARRANTY)
Firmware	not applicable

6. Related documents

- IG-TRF-DRY-630 — Installation and Commissioning Guide
- TG-TRF-DRY-630 — Troubleshooting and Error Code Guide
- MG-TRF-DRY-630 — Maintenance Guide
- FAQ-TRANSFORMER — Power transformers FAQ

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Cast Resin Dry-Type TRF 630kVA (TRF-DRY-630) — Installation and Commissioning Guide

1. Safety

Installation, wiring and commissioning must be performed by qualified personnel. Disconnect and lock out all power sources before work. Verify absence of voltage.

2. Receiving and storage

Inspect the packaging on arrival and report damage within 7 days (see POL-RETURN). Store indoors in a dry place between -25 and +55 °C, relative humidity below 85% without condensation.

3. Mounting and clearances

- Required clearances: Minimum 500 mm from enclosure walls; maintain electrical clearances per IEC 61936-1
- Operating temperature: -25 to +40 °C
- Degree of protection: IP00 (IP21/IP23 enclosure option)
- Above 1,000 m altitude, apply derating per the datasheet.

4. Wiring and tightening torques

- Terminal tightening torque: see dimension drawing
- Use copper conductors rated for at least 75 °C.
- Separate power and signal cables by at least 200 mm.

5. Commissioning procedure

1. Measure insulation resistance and polarisation index
2. Check off-circuit tap position against required secondary voltage
3. Verify temperature monitoring alarm and trip contacts
4. Check earthing of core and enclosure
5. Energise at no load and record secondary voltages

6. Pre-energisation checklist

☐ Product rating matches the drawing
☐ All terminals tightened to specified torque
☐ Protective earth connected
☐ Insulation resistance measured and recorded
☐ Settings / parameters recorded

7. Common commissioning issues

Symptom	Likely cause	See
Winding temperature alarm (140 °C)	Cooling fans not operating under high load	TG-TRF-DRY-630 §3.1
Abnormal humming noise	Loose core clamping or harmonic load	TG-TRF-DRY-630 §3.2
Low insulation resistance before energisation	Moisture absorption during storage	TG-TRF-DRY-630 §3.3
Secondary voltage too high	Off-circuit tap at wrong position	TG-TRF-DRY-630 §3.4

On-site commissioning support can be requested through the Global CS Center. For power transformers the request is handled by Field Service.

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Cast Resin Dry-Type TRF 630kVA (TRF-DRY-630) — Maintenance Guide

1. Maintenance intervals

Task	Interval
Visual inspection and cleaning	12 months
Insulation resistance	3 years
Temperature monitor function test	12 months
Oil DGA (oil-immersed)	12 months
Tightness of connections	12 months

2. Spare parts

- No user-replaceable parts; replace the complete unit.

3. End of life

Lifecycle status: ACTIVE. Expected service life is 20 years under normal conditions. Dispose of according to local regulations for electrical equipment.